ITCS216: Database Management Systems (1/2026)

Faculty of Information and Communication Technology

Workshop on Project Phase 1: Data Modeling

This workshop aims to provide a chance to refine the business rules and develop the data model for Project Phase I. All members of the team are required to analyze the business requirements (both rules and constraints) and sketch the data model using ERD Chen Notations.

Instructions:

1. **Use the group project to do an individual assignment.**
2. Starting from the M1 report, **each member** picks the different business process.
3. Revise / or create **at least 3 business rules** using **Template#1.**

Template#1: **Format:** Noun + Verb + Noun (e.g., a student enrolls many classes)

Note: you need to formulate both Backward and Forward

Business Process Name: Monthly Invoice		
Rule No. 1	**Rule Statement**	**Relationship**
Forward/Active	A member can receive multiple monthly invoices.	1: 1
Backward/Passive	Each monthly invoice belongs to only one member.	M :1
Conclusion	A member may have multiple monthly invoices.	1: 1

Business Process Name: Monthly Invoice		
Rule No. 2	**Rule Statement**	**Relationship**
Forward/Active	A monthly invoice can include multiple line items.	1:M
Backward/Passive	Each line item belongs to only one monthly invoice.	M:1
Conclusion	A monthly invoice may contain multiple line items.	1:M

Business Process Name: Monthly Invoice		
Rule No. 3	**Rule Statement**	**Relationship**
Forward/Active	An add-on service can be included in multiple monthly invoices.	1:M

Backward/Passive	A monthly invoice can include multiple add-on services.	1:M
Conclusion	Each add-on service can appear on multiple invoices, and each invoice can include multiple add-on services.	M:M

4. Revise / or create at least 2 business rules using **Template#2.**

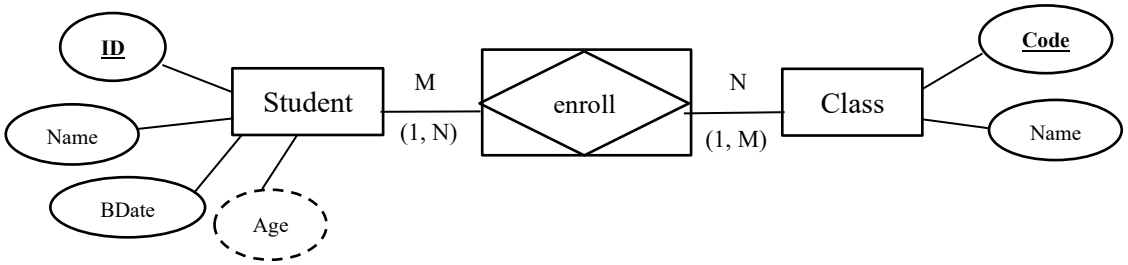
Template#2: **Format**: Noun + Verb + Attributes (e.g., a student comprises of ID, Name, Email),

Business Process Name: Monthly Invoice
Rule No. 1
A monthly invoice must have a unique Invoice_ID and include the issue date, payment due date, billing cycle, subtotal, tax rate, and total payable amount.

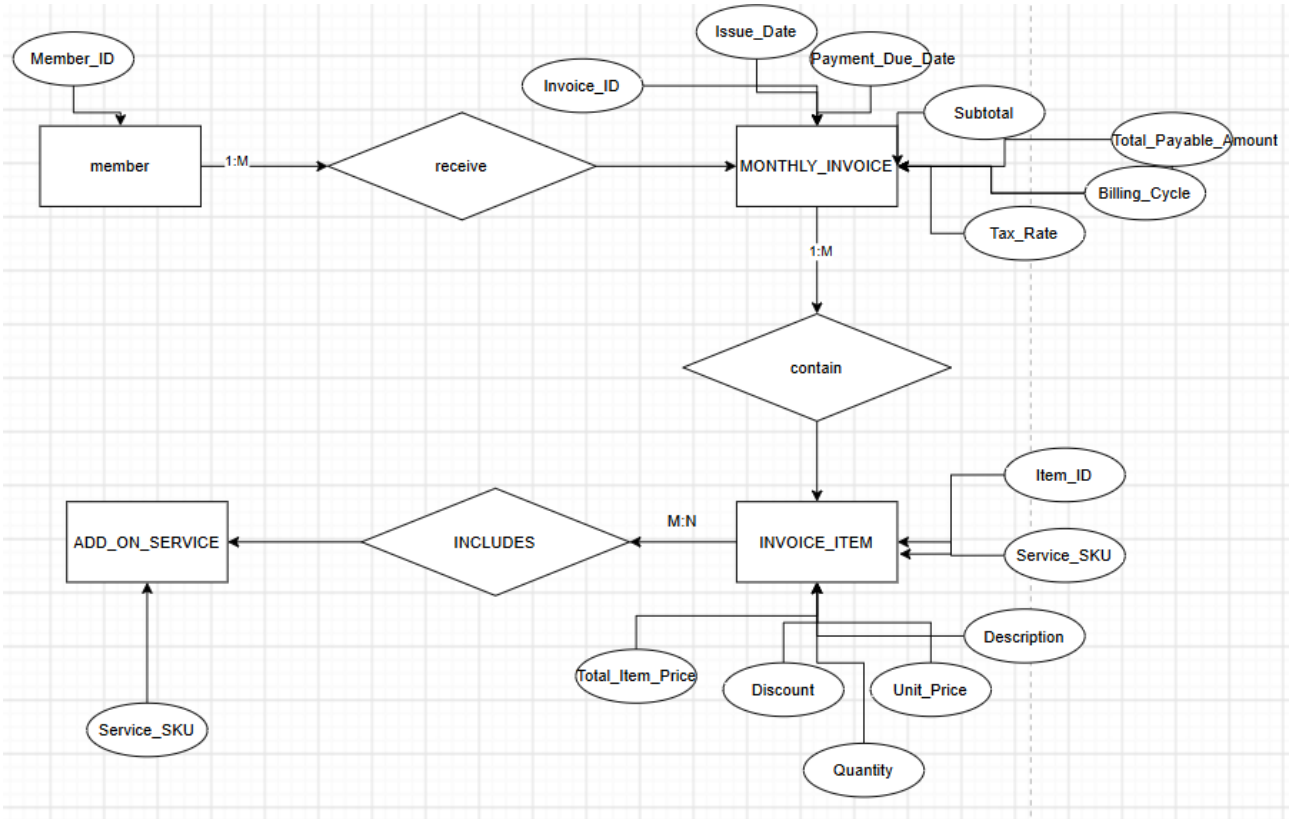
Business Process Name: Monthly Invoice
Rule No. 2
An invoice item must include Item_ID, Invoice_ID, service SKU, description, unit price, quantity, discount, and total item price.

5. Use the results obtained from Step 2-3 to draw the ERD (CHEN notation) using **Template #3**. All necessary notations for each rule have to demonstrate with complete and correct design

Template #3: Example

Business Process Name: Enrollment	
Rule No. 1	Students can enroll many classes. (M:N)
Rule No. 2	A student comprises of ID, Name, Birthdate, and calculated Age.
Rule No. X	A class consists of class code and name.
...	
ERD 	

6. Use the result from your teammate to create only one EERD diagram per group. (Recommended Tools: https://www.drawio.com/)


Member_ID
member
1:M
receive
Issue_Date
Payment_Due_Date
Invoice_ID
MONTHLY_INVOICE
Subtotal
Total_Payable_Amount
Billing_Cycle
Tax_Rate
1:M
contain
Item_ID
Service_SKU
INVOICE_ITEM
M:N
INCLUDES
ADD_ON_SERVICE
Service_SKU
Description
Total_Item_Price
Discount
Unit_Price
Quantity